

DAY 2 — AUGUST 28, LUGANO

Practical AI for Academics

Day 2 — hands-on with your own materials

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What do you actually need from AI?



Sophia Kazinnik  @SophiaKazinnik · 13h



I got *a lot* of questions on sophisticated workflows, agents, skills, orchestration, and increasingly elaborate setups (e.g., in the surveys I ran, the most requested topic by far was agents: "agent work," "agent building," "I want to learn how to work with AI agents.").

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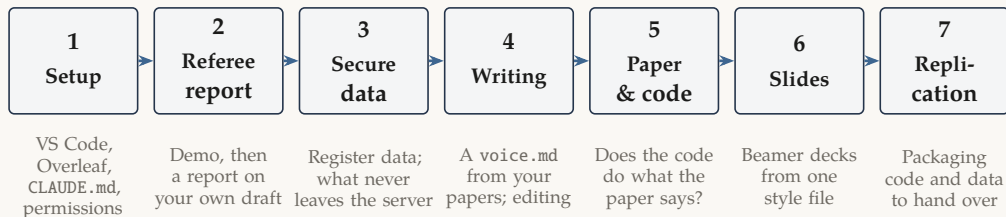


Lets be honest here. Most of our empirical research tasks are still fairly bounded: e.g., scrape data, clean data, run analysis, classify text, summarize documents, produce a forecast, repeat. You usually do not need a persistent autonomous agent, elaborate memory architecture, etc., to do those tasks well. In many cases, in fact, this is unproductive and adds complexity, cost, new failure points..

Key message: Context improves AI tools tremendously, and the goal is to provide the right context

A good workflow makes it easier to provide the useful context

Day 2 agenda — set up first, then hands-on with your own materials.



What you walk out with. A referee report on a draft you actually care about, a `voice.md` built from your own writing, and a report on whether your own code matches your own paper.

Blocks 2 and 4 are hands-on. Bring a draft you care about and two or three of your own papers.

We set the workspace up together in block 1, so nothing today assumes you finished the Day 1 setup.

Slides, skills, and guides for both days: claesbackman.com/lugano

SECTION 1

Setting up your workspace

Five steps get VS Code + Claude Code working in under five minutes.

1. Open VS Code and the Extensions panel (Ctrl+Shift+X / Cmd+Shift+X)
2. Search “**Claude Code**” and install *Claude Code for VS Code* (extension ID `anthropic.claude-code`)
3. Restart VS Code — Claude Code opens in the sidebar
4. Sign in with your Claude account (Pro/Max/Team/Enterprise all include Claude Code) or an API key
5. Open your **full project folder** (File > Open Folder) — never a single file

What an IDE actually is. Editor + file explorer + terminal + extensions, in one window. VS Code is the dominant free one. Think of it as Word or R Studio for code, LaTeX, markdown, and plain text.

How to get started

Get the skills from claesbackman.com/lugano — download the zip, or paste the one-line terminal command on that page. Source and updates: github.com/claesbackman/AI-research-feedback.

Unzip, then copy each skill folder (review-paper, review-paper-light, ...) into `~/.claude/skills/` — a hidden folder under your user folder (Windows: `C:\Users\you\.claude\skills`)

- To show hidden folders: Open File Explorer, click the View tab at the top, select Show, and check Hidden items
- A skill in `~/.claude/skills/` works in every project; one in `project/.claude/skills/` only in that project

Restart VS Code

The skill loads when you type `/skill-name`

A handful of extensions cover Stata, Python, LaTeX, and data

Stack	Extension ID	What it does
Stata	tmonk.stata-workbench	Run Stata from the editor
Python	ms-python.python	Syntax, IntelliSense, run button
LaTeX	james-yu.latex-workshop	Compile on save, side-by-side PDF preview, inline errors
Data	mechatroner.rainbow-csv	Colour-coded CSV columns; SQL-like filtering
	tomoki1207.pdf	Renders PDFs inline — read codebooks next to your code

KEY TAKEAWAY. *Install Python even if you do not write it. The agent uses Python as glue for many different tasks.*

You can link Overleaf to your own computer and work in that folder

Why bother? Overleaf has no Claude Code, no Codex, no Copilot, no proper diff view, no terminal. The fix: write in VS Code and let Overleaf follow what you produce.

Requirement. Both routes need **Overleaf Premium** — many universities provide it through the institutional login.

Every Claude Code session: open the folder, set context, then prompt.

1. Open VS Code and the project folder (File → Open Folder)
2. Open the Claude Code panel in the sidebar — or the terminal (Ctrl+`) and type `claude`
3. Confirm Claude is pointed at the **full project folder** — not a single file
4. Send a first orienting prompt

First prompt to try:

“Read the files in this folder and tell me what this project is about.”

Expected behaviour. Claude reads the folder, summarises the structure, and asks you what you want to do next.

CLAUDE.md is the standing-instructions file that is always loaded

About me

I am a researcher in [field]. I write papers, lecture notes, policy pieces.

How I want you to work with me

- Ask clarifying questions before generating long output.
- Keep a skeptical, critical tone in feedback. Do not flatter.
- When editing, preserve my voice. Avoid generic LLM phrasing.
- Avoid bullet lists and passive voice in formal writing.
- Cite the specific line or section you are commenting on.

Things to avoid

- Do not fabricate citations.
- Do not over-claim causality when summarising empirical work.
- Do not insert emoji or markdown decorations in formal documents.

A few files hold your durable project context

File	Use it for
<code>project/CLAUDE.md</code>	Project instructions. Dataset structure, sample restrictions, paper title, coauthor conventions, which script produces which output.
<code>~/ .claude/CLAUDE.md</code>	Personal preferences across every project: writing style, default tone, citation conventions. Project instructions are loaded later and usually carry more weight.
<code>voice.md</code>	Writing samples. Paragraphs, abstracts, and tone notes that Claude should match when editing prose.
<code>.claude/skills/...</code>	Reusable workflows. Named tasks you want to run again without rewriting the prompt.

KEY TAKEAWAY. *Keep secrets and raw confidential data **out of all of them** — these files are read into context on every session and may be sent to the model provider.*

Bootstrap CLAUDE.md in two minutes — type `/init`.

The built-in command: type `/init` in the Claude Code panel. Or spell it out:

“Read this project folder and draft a CLAUDE.md file with the project structure, conventions, and instructions you should follow. Do not edit anything else yet.”

What Claude does

- Reads the folder
- Infers languages, structure, conventions
- Generates a CLAUDE.md draft

Then you edit it. Add tone preferences, things to avoid, sample restrictions Claude could not infer.

Permissions follow a tiered model

Plan mode

Reads and proposes; cannot write.

Manual

Asks before each edit or command.

Edit automatically

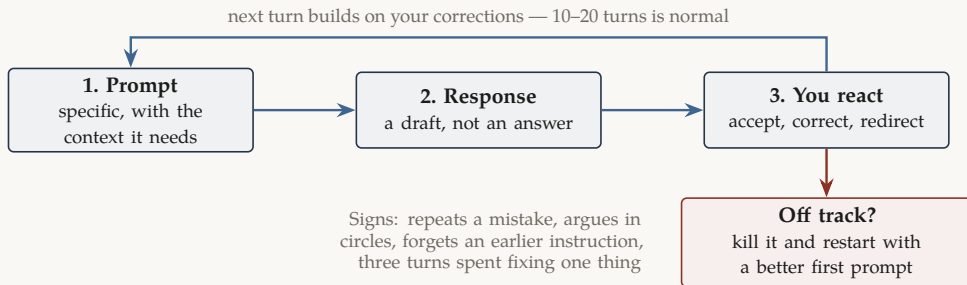
Claude will edit your selected text or the whole file

Auto mode (default mode now)

Claude will approve actions that pass a safety check and pause for anything risky

Claude Code should stay inside the folder you opened, but this is not certain. Use `/permissions` to see the current rules, and allow- or block-list specific commands (`Rscript`, `pdflatex`, `rm`) in `settings.json`.

A session is a conversation, not a query



Continue the conversation. The context carries your corrections, so later answers improve: *“Shorter, and use the Danish case.” “You misread Table 2 — look again.” “Now the same for Section 3.”*

Kill it when you veer off track. Start fresh and put what you learned into the first prompt — or into `CLAUDE.md` so you never repeat it.

SECTION 2

A referee report on your paper

*Pick one draft you would like another pair of eyes on.
Preferably a relatively close-to-finished draft.*

Then write /review-paper

Live demo — /review-paper on a real draft, eight subagents in parallel.

Setup

- One of my own papers, in its own folder
- .tex source, .bib, supporting materials
- One command: /review-paper with a target journal

Output

- A structured markdown report
- Referee-style comments, grouped by theme
- Specific line and section references

Eight subagents running in parallel

- Spelling, grammar & style
- Internal consistency & cross-references
- Unsupported claims & identification
- Mathematics, equations & notation
- Tables, figures & documentation
- Referee assessment: positioning & fit
- Contribution advocate
- Contribution skeptic

Pattern: eight narrow critics beat one diffuse one.

KEY TAKEAWAY. *The skill is a ~200-line markdown file — read it, edit it, download it.*

Hands-on — run a referee report on your own writing.

1. Open one of your own drafts in VS Code
2. Make sure it is in markdown or LaTeX — convert if needed
3. Run `/review-paper-light` for a 1–2 minute version, or `/review-paper` for the full eight-agent run
4. Read the output. It's not all correct!

Debrief questions

- What did the report catch that you would not have seen by yourself?
- What did it get wrong?
- Which subagent gave you the most useful feedback?

A skill is a markdown file the agent loads on demand — invoke with /name.

A skill lives at `.claude/skills/your-skill/SKILL.md` (or `~/.claude/skills/` for every project):

```
---
name: review-paper
description: Run an 8-agent pre-submission referee report for an academic paper
targeting a specified journal
---
You are coordinating a rigorous pre-submission review of an
academic economics paper. You will run 8 specialized review agents
in parallel and consolidate their findings into a structured report.
```

Invoke with `/review-paper`. Claude loads the instructions and runs them.

CLAUDE.md is always loaded, skills are on demand.

CLAUDE.md — always loaded

- Loaded into context every session
- Background, conventions, voice
- “How I want you to work with me”

Cost: permanent context budget.

Skill — on demand

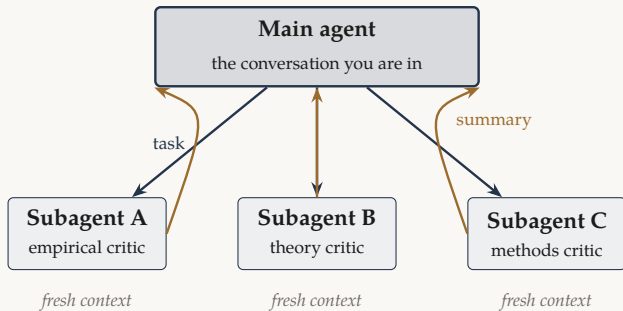
- Loaded only when invoked with /name
- A specific workflow, with structured output
- “Do this thing for me”

Cost: zero until triggered.

KEY TAKEAWAY. *If you find yourself pasting the same prompt multiple times, you can ask Claude to make it into a skill.*

*Now launch another agent to critique the referee report
you just generated — ask Claude to write the
instructions.*

Subagents add two things the main agent cannot do alone — parallelism and context isolation.



KEY TAKEAWAY. *The /review-paper skill spawns **eight** subagents.*

The second-opinion pattern — spawn an adversary to check on the work.

The main agent sees its own prior answer and has reason to defend it.

Why a fresh subagent is different

- No chat history: no anchoring on what you already wrote
- No conversational momentum: no reason to agree with the main agent
- Clean context: can hold the whole artifact at once

When to spawn one

- Before accepting a long answer: *“is this actually right?”*
- After a long refactor: *“what did we break?”*
- When you suspect flattery: *“give me the honest version”*
- Before sending: *“read this as a hostile reader”*
- After doing automatic edits: *“look over the changes we did”*

KEY TAKEAWAY. The phrases **“spawn a subagent”** or **“launch an agent”** are the trigger words. The main agent knows what to do and will write the instructions for you.

What the demo is meant to show

1. **Grounded output.** The report is about *your* draft, not a generic checklist
2. **Subagents are why this works.** Eight narrow critics catch things one diffuse reader misses
3. **Not a replacement for a referee.** It is a pre-flight check before you submit
4. **Transparent.** The skill is a ~200-line markdown file — fork it, modify it, make it yours

KEY TAKEAWAY. *The point is not to trust the report. The point is to make hard feedback cheap enough that you ask for it before submitting — not after the desk reject.*

Other tools generate referee reports too

Tool	What it does	Pricing
Refine.ink	Referee-style report on a paper PDF	<i>Paid</i>
coarse.ink	Lightweight version of the same idea	<i>≈ free</i>
Referee 2	Paid referee-report tool	<i>Paid</i>
Stanford Agentic Reviewer	Open agentic reviewer	<i>Free</i>

KEY TAKEAWAY. *The advantage of running your own skill is that it lives next to your draft, knows your conventions, and you can edit what it checks.*

SECTION 3

Working with secure data

You should never paste any sensitive data or results into Claude

Files or chats that Claude read go to Anthropic servers

Confidentiality and data

- **Never paste** confidential microdata, register data, or unpublished co-authored work.
- **“Training on your data.”** Some consumer tiers may use your chats by default. Know the setting; prefer business/team tiers for anything sensitive.
- **When in doubt, abstract it.** Ask about a made-up example or use a synthetic dataset.
- **Check institutional policy** before putting any work material into any AI tool.

Secure servers split the workflow

What Claude can still help with on your laptop

- Your .do files
- Writing and iterating on empirical strategy

Claude cannot run the files or read the logs on the server, but can still be very helpful in a number of ways

Two artifacts to keep local

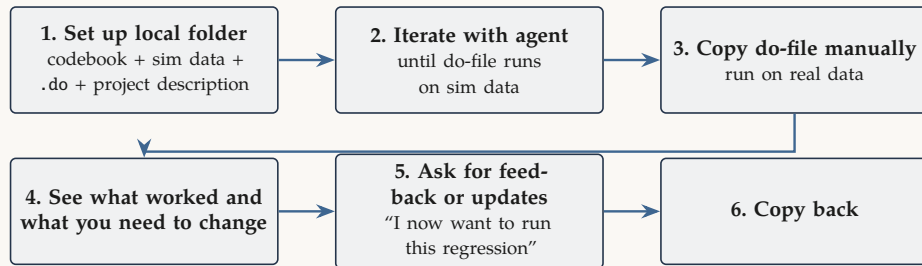
1 Codebook in markdown

- Convert once (/pdf-to-markdown)
- Variable names, labels, value ranges, missing-value codes
- Agent reads it before writing recode, label define, merge

2 Simulated dataset

- Same variable names, types, ranges — not real values
- AER/JF/ReStud replication packages often ship with these
- Or generate from the codebook

A workflow for secure data on servers



KEY TAKEAWAY. *The agent never sees the real data (and you should never paste any result from the server into Claude)—but it sees everything that explains the data. That is usually enough to write correct Stata code.*

Does step 2 actually work? Tested on a Danish register codebook

Artifact 2 above says “or generate from the codebook”. Here is what that produces.

Two codebooks (BEF population, IND income), no data, fresh session. The agent produced 55,000 rows per register with:

- Correct variable names — PNR, KOEN, CIVST, ERHVERVSINDK_13, FORM, QSPLINDK
- Correct value codes — KOEN in {1,2}, CIVST in {U,G,F,E}, IE_TYPE in {1,2,3}
- A working PNR–AAR merge key across both registers

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KEY TAKEAWAY. Code written against this runs unchanged on the real registers. The schema is right, so merges, reshapes, variable names and sign conventions are all tested before you ever enter the secure environment.

What synthetic data does and does not buy you

Tests that the code RUNS

Variable names, merge keys, reshapes, type handling, sign conventions, the shape of the panel. All the things that produce an error message. This is genuinely most of the debugging.

Does NOT test that it is RIGHT

Correlations, magnitudes, selection, missingness patterns. The fake data can only contain what someone thought to put in it — so it cannot catch a bug nobody imagined.

KEY TAKEAWAY. Use it to arrive at the secure environment with working code. Not to arrive with verified code.

How to make this safe(r)?

1. **Make the agent confess.** “List every assumption the codebook did not give you.”
The list is the specification of what you still have to check — and it is free.
2. **Reproduce one known number first.** Inside the secure environment, before anything new: rebuild a published figure, a table from an earlier paper, an official statistic.
This is the only step that catches a silently wrong sample.

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You need to verify the output.

SECTION 4

Writing with agentic AI

Most of the value is not in having AI write your text

I like writing, and I don't want AI to take away part of the job that I like

But: I am not a native speaker, and I make mistakes

AI can be very helpful in helping me structure ideas or fix mistakes, but it is worth emphasizing that a lot of AI writing is bad.

Use it for what it is good for, and learn to ignore some of what it says

A `voice.md` file makes edits sound like you

When editing my prose, match the voice in `voice.md`.
Do not use em-dashes, bullet lists, or passive constructions unless the original does.
Do not use "it's not x, it's y"-style writing.

What goes in it

- 5–10k words of your prose — intros, abstracts, paragraphs you are proud of

KEY TAKEAWAY. *Reference it from `CLAUDE.md` to give Claude writing context for every session*

Hands-on (15 min) — build a `voice.md` from your own writing.

We use a skill written by Mihail Velikov: `voice-extractor`.

1. Upload 4 PDFs of your own writing to a folder
2. Call the skill: `/voice-extractor sample-papers`
3. Read through `voice.md` and verify — edit anything that does not sound like you

What goes in. Intros, abstracts, paragraphs you are proud of. 5–10k words is plenty.

What stays out. Co-authored sections where the prose is not yours. AI-edited drafts.

KEY TAKEAWAY. Reference `voice.md` from `CLAUDE.md`. Every future session loads your voice into context.

/prose-revision fixes the order of the ideas before the choice of words.

What comes back. A diagnosis of the idea flow first, then two or three rewrites that differ in how far they restructure, and a short note on the tradeoff between them. Claims, citations, and notation come through unchanged.

Where it stops. It will not add or drop an idea without saying so, and it is not a proofreader. Revision comes first, the typo pass comes last.

From https://x.com/alz_zyd_

AI detectors can tell AI generated writing

How they work. A classifier trained on the fingerprint of model text — predictable word choice, even rhythm, stock phrasing. It returns a probability, not evidence.

Pangram. The most popular current detector, with very low reported false-positive rates.

SECTION 5

Verifying the paper-code link and changes to the code

/review-paper-code asks whether the code does what the paper says.

What it does, in four steps

1. Finds your main `.tex` and your code folder on its own — Stata, R, Python
2. Reads the paper for the sample, the specification, and the headline numbers
3. Runs two agents at once: reproducibility, and paper-to-code mapping
4. Reconciles the two and writes `reviews/code_review_report.md`

Call it: `/review-paper-code`, adding `full` to review every file it finds.

What comes back

- **Reproducibility checklist:** paths, seeds, ungenerated outputs, dependencies, run order
- **Paper-code map:** matched, worth verifying, or not found in the files reviewed
- **Likely discrepancies:** where the two disagree
- **Next steps,** ordered by importance

KEY TAKEAWAY. Does Table 3 come from the regression the paper describes? Cheapest to answer before the replication package goes out.

A skill: `/explain-diff`

`/explain-diff` checks what changed within a Github repo and generates a html site that you can open

- **Verification.** It diffs generated tables, compares N and headline coefficients across logs, and quotes numbers only from output files it has opened.
- **Output.** One offline HTML page in five sections: what the code did before, what changed and why, consequences for the results, a code walkthrough grouped by purpose, and a five-question quiz. Saved outside the repo with a plain-text summary in the chat.

Dropbox or Github?

	GitHub	Dropbox
Sync	Manual push and pull	Automatic, within seconds
History	Full git history, branches	Dropbox revisions only
Best for	Solo work, versioned drafts	Live coauthoring in Overleaf
Conflicts	Git merge	“Conflicted copy” files

KEY TAKEAWAY. ***Pick one per project.** Both syncs at once means duplicate-file chaos.*

Two routes: GitHub Desktop for a paper trail (click to commit, no terminal), Dropbox for live coauthoring.

Setup A GitHub via GitHub Desktop

1. Overleaf: Menu → Integrations → GitHub → **Export Project to GitHub**
2. GitHub Desktop: File → Clone repository, pick the repo and a local folder
3. Open that folder in VS Code
4. Edit in VS Code. In GitHub Desktop: one-line summary, **Commit to main, Push origin**
5. Overleaf: Menu → GitHub → **Pull GitHub changes into Overleaf**
6. Coauthor edited in Overleaf? **Push Overleaf changes to GitHub**, then pull in GitHub Desktop

Accept the TeX `.gitignore` template: no `.pdf` or `.aux` in the history.

Setup B Dropbox

1. Install Dropbox and let the initial sync finish
2. Overleaf: Menu → Dropbox Sync → **Link**. Overleaf creates `Apps/Overleaf/` with one subfolder per project
3. Open the project folder in VS Code and edit as normal — every save reaches Overleaf within seconds, and Overleaf edits flow back
4. Compile in Overleaf as usual, or locally with `latexmk -pdf`

No commit step. Good for active coauthoring, less good if you want a paper trail.

SECTION 6

Slides

Making slides with AI

The old path

- Open Beamer
- Write LaTeX by hand
- Fight the syntax
- Iterate slowly

Hours of mechanical work between idea and first draft.

The Claude Code path

- Open the project folder
- Hand Claude a file: “*Make a Beamer deck based on @paper.tex*”
- Compile, look at the output, iterate

This deck was built that way.

KEY TAKEAWAY. *This also means that we have to practice presentations in a different way*

Including a style file gives Claude guidance to match the look you want.

Without a style file. Claude defaults to generic Beamer themes.

With a style file. The deck inherits your colour palette, frame structure, and title conventions.

How to build a `slides-style.md`

- Ask Claude to analyze your old presentations or to make up something that looks good
- Save the result as `slides-style.md` in the project folder
- Reference it whenever you ask for a new deck

KEY TAKEAWAY. *The Day 1 deck, this deck, and your future decks can all share one style file. Build it once.*

SECTION 7

A replication package

A replication package is a README, a master script, and portable paths.

What has to be in the folder

1. A README: data sources, access terms, software versions, runtime
2. A table mapping each exhibit to the script that makes it and the file it writes
3. One master script that runs the pipeline in order, relative paths, seeds set
4. The data you may ship, and access instructions for the data you may not

What to ask the agent

- “Read Code/ and draft the README, including the exhibit-to-script table”
- “List every input file the code reads; flag the ones not in this folder”
- “Make every path relative and write `master.do`”

Then verify: run it in a fresh folder and diff the tables against the paper.

KEY TAKEAWAY. *With register data the code and the README are the package — the data never leaves the server.*

SECTION 8

Conclusions

Key message: Context improves AI tools tremendously, and the goal is to provide the right context

Key difference: how you provide the useful context and how easy it is

Q & A

Open floor. Email: claes.backman@gmail.com |
Slides, skills, guides: claesbackman.com/lugano

References I

Appendix — CLAUDE.md starter template.

About me

I am a researcher in [field].

How I want you to work with me

- Ask clarifying questions before generating long output.
- Critical, skeptical tone in feedback. Do not flatter.
- When editing, preserve my voice (see voice.md). No generic LLM phrasing.
- Avoid bullet lists and passive voice in formal writing.
- Cite the specific line or section you are commenting on.

Things to avoid

- Do not fabricate citations.
- Do not over-claim causality.
- Do not insert emoji or markdown decorations in formal docs.

Appendix — instructor's skill library.

Skill	Use
<code>/review-paper</code>	Pre-submission referee report, 8 agents
<code>/review-paper-light</code>	Fast 2-agent version, $\approx$ 1 minute
<code>/review-paper-code</code>	Paper + code alignment review
<code>/review-grant</code>	Pre-submission grant review, 6 agents
<code>/review-pap</code>	Pre-analysis-plan review
<code>/audit-analysis</code>	Adversarial audit of changed analysis code
<code>/explain-diff</code>	Explain a code change as an HTML page with a quiz
<code>/paper-version</code>	Paper $\rightarrow$ policy brief / 1-page / 5-page
<code>/pdf-to-markdown</code>	Convert PDFs to readable markdown

All at github.com/claesbackman/ai-research-feedback (MIT), linked from claesbackman.com/lugano.
Fork them, modify them, make them yours.

Appendix — file format quick reference.

Format	Handling
.md, .tex, .txt, .bib	Native — no conversion
.py, .R, .do, .ipynb	Native — read, edit, run
.csv	Native up to a few thousand rows
.docx	Convert: <code>pandoc file.docx -o file.md</code>
.xlsx	Export to CSV first
.dta, .rds	Script-based conversion to CSV
PDF (born-digital, short)	Native; expect table mangling
PDF (long, scanned, equations)	Convert to markdown first

Appendix — signs of session drift.

- Claude ignores `CLAUDE.md`
- Repeats mistakes you already corrected
- Invents files that do not exist
- Gives circular answers
- Forgets the last three turns

Fix. Start a fresh session or close and reopen. Re-state the task. Re-attach the relevant file.

30+ turns or the task has shifted → start fresh. Always.